

# Cell Biology Mock Exam and Answer Key

2026-04-27

## Exam

### Instructions

- This mock exam is designed to roughly match the style and difficulty of 2026CB\_homework\_I.pdf.
- It aims to cover about 70–80% of the major knowledge points in the current notebook and lecture PDFs.
- Total score: 102.

### I. Multiple choice questions

Questions 1–62, 1 point each. Select the best answer unless otherwise instructed.

1. Which of the following is the best example of a prokaryotic homolog of tubulin?
  - a. MreB
  - b. ParM
  - c. FtsZ
  - d. Crescentin
2. Which cytoskeletal system is most directly responsible for the contractile ring during cytokinesis?
  - a. Actin filaments with myosin II
  - b. Intermediate filaments
  - c. Microtubules with dynein
  - d. Centrosomal lamins

3. Which cytoskeletal element is most important for long-distance intracellular transport of membrane-enclosed organelles?
  - a. Actin cortex
  - b. Microtubules
  - c. Intermediate filaments
  - d. Desmosomes
4. Stress fibers are best described as:
  - a. Branched actin networks under the plasma membrane
  - b. Contractile actin bundles in non-muscle cells
  - c. Intermediate filament cables in epithelia
  - d. Microtubule bundles in migrating cells
5. Which protein complex primarily nucleates branched actin networks?
  - a. Formin
  - b. Arp2/3 complex
  - c. Dynactin
  - d. Katanin
6. Which protein is most directly associated with elongation of long, unbranched actin filaments?
  - a. Formin
  - b. Arp2/3
  - c. CapZ
  - d. Cofilin
7. CapZ primarily acts by:
  - a. Severing actin filaments
  - b. Binding ATP-actin monomers
  - c. Capping the plus end of actin filaments
  - d. Linking dynein to membranes
8. Cofilin promotes actin filament turnover mainly by:

- a. Capping the minus end
  - b. Twisting and destabilizing the filament
  - c. Converting ADP to ATP on actin
  - d. Anchoring actin to Z discs
9. The motor activity of myosin II is most closely coupled to:
- a. GTP hydrolysis
  - b. ATP hydrolysis and Pi release
  - c. ADP synthesis
  - d.  $\text{Ca}^{2+}$  export
10. In skeletal muscle, the plus ends of thin filaments are anchored at the:
- a. M line
  - b. A band
  - c. Z disc
  - d. H zone
11. Which protein acts as a molecular spring in the sarcomere?
- a. Tropomodulin
  - b. Titin
  - c. CapZ
  - d. Dynein
12. Smooth muscle contraction is directly activated when  $\text{Ca}^{2+}$ :
- a. Binds troponin C on actin
  - b. Binds calmodulin and activates MLCK
  - c. Causes CapZ dissociation
  - d. Stabilizes tropomyosin on microtubules
13. Dynamic instability of microtubules depends primarily on:
- a. ATP hydrolysis on  $\alpha$ -tubulin
  - b. GTP hydrolysis on  $\beta$ -tubulin
  - c. Dynein detachment

- d. Actin treadmilling
14. Loss of the GTP cap from a microtubule plus end usually triggers:
- a. Rescue
  - b. Treadmilling
  - c. Catastrophe
  - d. Annealing
15. The major function of  $\gamma$ -tubulin is to:
- a. Cap actin filaments
  - b. Nucleate microtubules
  - c. Stabilize intermediate filaments
  - d. Carry cargo to the Golgi
16. Which protein severs microtubules?
- a. Katanin
  - b. Tropomodulin
  - c. Lis1
  - d. Ankyrin
17. Most kinesins move toward the:
- a. Minus end of actin filaments
  - b. Plus end of microtubules
  - c. Minus end of microtubules
  - d. Centrosomal matrix
18. Dynein attachment to membrane-enclosed organelles is commonly mediated by:
- a. CENP-A
  - b. Dynactin
  - c. H1
  - d. SRP
19. In motile cilia and flagella, dynein-driven sliding between adjacent doublets is converted to bending by:

- a. DNA looping
  - b. Flexible linker proteins
  - c. ATP synthase
  - d. Capping proteins
20. Plectin is best described as a protein that:
- a. Generates ATP in mitochondria
  - b. Cross-links intermediate filaments to other cytoskeletal systems
  - c. Caps the plus end of actin filaments
  - d. Forms the nuclear pore basket
21. Members of the nuclear lamina structural network belong to which family?
- a. Type V intermediate filaments
  - b. Actin-related proteins
  - c. Kinesin family proteins
  - d. Histone variants
22. Which one is an A-type lamin?
- a. Lamin B1
  - b. Lamin B2
  - c. Lamin C
  - d. CENP-A
23. Karyotyping is most useful for detecting:
- a. Point mutations in coding regions
  - b. Single amino acid substitutions
  - c. Aneuploidy and large chromosomal rearrangements
  - d. mRNA splicing defects
24. Which statement about nucleosome positioning is most accurate?
- a. DNA sequence has no influence on nucleosome formation
  - b. Only histone H1 determines nucleosome location
  - c. DNA bendability contributes to preferred nucleosome positions

- d. Nucleosomes form only on repetitive DNA
25. Linker histone H1 primarily:
- a. Replaces histone H3 in centromeres
  - b. Changes DNA exit path and promotes chromatin compaction
  - c. Catalyzes DNA methylation
  - d. Binds directly to RNA polymerase II CTD
26. ATP-dependent chromatin remodeling complexes can:
- a. Translate mRNA into protein
  - b. Slide, remove, or exchange nucleosomes
  - c. Hydrolyze histones into peptides
  - d. Ligate Okazaki fragments
27. The histone variant that helps define many centromeres is:
- a. H1
  - b. HP1
  - c. CENP-A
  - d. Xist
28. Self-propagation of heterochromatin is best explained by:
- a. Random diffusion of histones
  - b. Reader-writer positive feedback
  - c. Ribosome assembly
  - d. GPI anchoring
29. Which method gives a population-averaged contact map of chromosome conformation?
- a. FRAP
  - b. 3C/Hi-C
  - c. FRET
  - d. Patch clamp
30. Chromosome tracing is especially valuable because it:

- a. Measures ATP turnover in chromatin
  - b. Directly visualizes chromosome organization in single cells
  - c. Replaces all sequencing-based methods
  - d. Detects only coding sequences
31. In polytene chromosomes, the chromocenter forms from:
- a. Mitochondrial DNA clusters
  - b. Aggregated heterochromatin near centromeres
  - c. Fused nucleoli
  - d. Centrosomal satellites
32. Gene duplication is evolutionarily important because it:
- a. Prevents any future mutation
  - b. Provides raw material for new gene functions
  - c. Eliminates homologous genes
  - d. Always reduces genome size
33. Which RNA polymerase mainly transcribes mRNAs?
- a. Pol I
  - b. Pol II
  - c. Pol III
  - d. Pol IV
34. Which RNA polymerase mainly transcribes tRNAs and 5S rRNA?
- a. Pol I
  - b. Pol II
  - c. Pol III
  - d. Telomerase
35. DNA superhelical tension generated during transcription is relieved by:
- a. Topoisomerases
  - b. Restriction enzymes
  - c. SRP

- d. Cohesin
36. The earliest processing event on a eukaryotic pre-mRNA is:
- a. Polyadenylation
  - b. 5' capping
  - c. Spliceosome disassembly
  - d. tRNA charging
37. Formation of the 3' end of most eukaryotic mRNAs requires:
- a. RNA editing only
  - b. Direct transcription termination with no processing
  - c. Cleavage followed by poly(A) addition
  - d. Reverse transcription
38. DNA looping in gene regulation is important because it:
- a. Prevents all enhancer action
  - b. Allows distant regulators to contact promoter-bound machinery
  - c. Converts DNA into RNA
  - d. Forces nucleosomes off the chromosome permanently
39. Transcriptional synergy refers to:
- a. Equal contribution of all genes
  - b. A less-than-additive effect of two activators
  - c. A greater-than-additive effect of multiple activators
  - d. Inhibition of one activator by another
40. Insulator DNA sequences primarily function to:
- a. Recruit ribosomes
  - b. Block inappropriate long-range gene regulation
  - c. Replicate telomeres
  - d. Polymerize actin
41. A major molecular basis for cell memory is:



- a. Positive feedback and heritable chromatin states
  - b. Loss of all transcription factors after mitosis
  - c. Removal of all DNA methylation
  - d. Complete absence of signaling pathways
42. Genomic imprinting is best defined as:
- a. Expression of both alleles equally
  - b. Parent-of-origin-dependent monoallelic expression
  - c. Random loss of one chromosome
  - d. Universal repression of maternal genes
43. X-chromosome inactivation in mammals is most closely associated with:
- a. CapZ
  - b. Xist RNA
  - c. Tropomodulin
  - d. SRP54
44. Alternative cleavage and polyadenylation (APA) can change:
- a. Only the DNA replication origin
  - b. The fate and coding output of the transcript
  - c. Ribosome structure
  - d. The polarity of actin filaments
45. The most abundant phospholipid in many eukaryotic membranes is:
- a. Phosphatidylcholine
  - b. Cardiolipin
  - c. Phosphatidylinositol trisphosphate
  - d. Cholesteryl ester
46. Cholesterol helps membranes by:
- a. Hydrolyzing phospholipids
  - b. Filling gaps created by unsaturated hydrocarbon tails
  - c. Removing all membrane asymmetry

- d. Serving as a motor protein
47. Which membrane motion is rare without enzymes?
- a. Lateral diffusion
  - b. Rotation
  - c. Flexion
  - d. Flip-flop
48. Which class of membrane proteins usually requires detergent for solubilization?
- a. Peripheral membrane proteins
  - b. Integral membrane proteins
  - c. Histones
  - d. Ribosomal proteins
49. Hydropathy plots are most useful for predicting:
- a. DNA methylation islands
  - b. Potential membrane-spanning hydrophobic segments
  - c. RNA splicing enhancers
  - d. Chromosome territories
50. Glycosylation of membrane proteins generally faces the:
- a. Cytosolic side
  - b. Non-cytosolic/extracellular side
  - c. Centrosomal side
  - d. Nucleoplasmic side
51. Which of the following is NOT one of the three major membrane-bending mechanisms emphasized in class?
- a. Hydrophobic insertion
  - b. Scaffolding
  - c. Protein crowding
  - d. DNA ligation
52. Transporters differ from channels mainly because transporters:

- a. Never bind cargo
  - b. Undergo alternating-access conformational changes
  - c. Are always faster than channels
  - d. Can move only water
53. The three classic states of a transporter are:
- a. Bound, unbound, degraded
  - b. Open, closed, cleaved
  - c. Outward-open, occluded, inward-open
  - d. Resting, spiking, refractory
54. Which of the following is NOT a major class of ATP-driven pump?
- a. P-type
  - b. ABC
  - c. F-type/V-type
  - d. Arp-type
55. Bacterial cells protect themselves against extreme osmotic shock using:
- a. Mechanosensitive channels
  - b. Histone H1
  - c. Nuclear pores
  - d. Tight junctions
56. The nicotinic acetylcholine receptor at the neuromuscular junction is:
- a. A GPCR
  - b. A voltage-gated  $K^+$  channel
  - c. A ligand-gated cation channel
  - d. A proton pump
57. Long-term potentiation in the hippocampus depends critically on:
- a.  $Ca^{2+}$  entry through NMDA receptors
  - b. ATP export through VDAC
  - c. Histone H1 phosphorylation

- d. Dynein-driven vesicle bending
58. Which transport mechanism moves folded proteins between nucleus and cytosol?
- a. Vesicular transport
  - b. Gated transport
  - c. Transmembrane translocation through TOM
  - d. Endocytosis
59. The outer nuclear membrane is continuous with the:
- a. Lysosomal membrane
  - b. ER membrane
  - c. Plasma membrane
  - d. Mitochondrial inner membrane
60. Most proteins imported into mitochondria enter first through:
- a. Sec61
  - b. TOM complex
  - c. NPC
  - d. COPII
61. A classic peroxisomal targeting signal is:
- a. KDEL at the N-terminus
  - b. SKL at the C-terminus
  - c. NLS rich in lysine
  - d. A transmembrane  $\beta$ -barrel
62. Which statement best describes the role of SRP?
- a. It splices introns from mRNA
  - b. It recognizes ER signal peptides and pauses translation transiently
  - c. It degrades misfolded proteins in lysosomes
  - d. It methylates histones at centromeres

## II. Short-answer questions

Questions 1–8, 5 points each.

1. A dividing cell is treated with nocodazole, while another is treated with Taxol. Both populations stop proliferating. Explain why these two drugs, despite having opposite effects on microtubule polymer mass, can both block mitosis.
2. Compare branched actin networks and actin bundles in terms of nucleating proteins, architecture, and typical cellular functions.
3. Explain why membrane asymmetry is functionally important, and why the orientation of membrane proteins and lipids is preserved during vesicular transport.
4. A mutation disrupts the reader protein in a heterochromatin spreading system. Predict the effect on heterochromatin propagation and explain the logic using the reader-writer model.
5. Explain how Ran GTPase imposes directionality on nuclear import and export.
6. A membrane protein is correctly synthesized but persistently misfolds in the ER. Describe the pathway that removes it and explain why this pathway is important for cell physiology.
7. Compare 3C/Hi-C and chromosome tracing as methods for studying 3D genome organization. What does each method measure best, and why are they complementary?
8. Explain how alternative cleavage and polyadenylation could allow the same gene to produce a membrane-bound form in one cell type and a secreted form in another.

## Answer Key

### I. Multiple choice answers

1. c
2. a
3. b
4. b
5. b
6. a
7. c
8. b
9. b
10. c
11. b
12. b
13. b
14. c
15. b
16. a
17. b
18. b
19. b
20. b
21. a
22. c
23. c

24. c

25. b

26. b

27. c

28. b

29. b

30. b

31. b

32. b

33. c

34. c

35. a

36. b

37. c

38. b

39. c

40. b

41. a

42. b

43. b

44. b

45. a

46. b

47. d

48. b

- 49. b
- 50. d
- 51. d
- 52. b
- 53. c
- 54. a
- 55. c
- 56. a
- 57. b
- 58. b
- 59. b
- 60. b
- 61. b
- 62. b

## II. Short-answer reference answers

1. **Nocodazole** promotes microtubule depolymerization, whereas **Taxol** stabilizes microtubules and suppresses their dynamic instability. Mitosis requires highly dynamic spindle microtubules to search for kinetochores, capture chromosomes, align them, and then reorganize during segregation. Therefore, both too little microtubule polymer and overly stabilized, non-dynamic microtubules disrupt spindle function and block mitosis.
2. Branched actin networks are nucleated mainly by the **Arp2/3 complex**, which generates daughter filaments from the sides of preexisting filaments, producing dense dendritic arrays. These networks are typical of lamellipodia and the cell cortex, where they help drive membrane protrusion and rapid shape change. In contrast, actin bundles are commonly built from long, relatively straight filaments promoted by **formin**; they are then organized by cross-linking proteins into parallel or contractile bundles. Bundles function in structures such as microvilli, filopodia, stress fibers, and the contractile ring.



3. Membrane asymmetry is important because the two leaflets and the two faces of membrane proteins perform different functions. For example, glycolipids and many glycosylated domains face the non-cytosolic/extracellular side, whereas many signaling and cytoskeletal interactions occur on the cytosolic side. During vesicular transport, membrane topology is preserved: the lumen of the ER/Golgi corresponds topologically to the extracellular space, so proteins and lipids do not flip their orientation when they move in vesicles.
4. In the reader-writer model, the **reader** recognizes an existing heterochromatin mark and recruits the **writer**, which deposits the same mark on adjacent nucleosomes. If the reader is lost, the system can no longer efficiently recognize the existing modification and spread it outward. As a result, heterochromatin propagation is impaired or stops, and the self-propagating maintenance of the silent state becomes unstable.
5. Ran directionality depends on compartment-specific regulation of its nucleotide state. In the nucleus, **Ran-GEF** generates high **Ran-GTP**; in the cytosol, **Ran-GAP** promotes Ran-GTP hydrolysis to Ran-GDP. During import, import receptors bind cargo in the cytosol, pass through the nuclear pore, and release cargo when they encounter Ran-GTP in the nucleus. During export, export receptors bind cargo together with Ran-GTP in the nucleus, carry the complex out, and then release cargo after GTP hydrolysis in the cytosol. This asymmetric Ran system imposes directionality on both import and export.
6. Persistently misfolded ER proteins are removed by **ER-associated degradation (ERAD)**. The protein is recognized by the ER quality-control system, retrotranslocated or dislocated back across the ER membrane, ubiquitinated, extracted by ATP-dependent factors, and degraded by the proteasome in the cytosol. ERAD is important because it prevents toxic accumulation of misfolded proteins, maintains ER homeostasis, and helps limit ER stress.
7. **3C/Hi-C** measures contact frequencies between chromosomal loci across populations of cells, making it especially powerful for identifying average features such as compartments, TAD-like organization, and long-range interaction tendencies. **Chromosome tracing** uses imaging to visualize chromosome folding directly in single cells, so it is best for revealing cell-to-cell variability and actual spatial configurations. They are complementary because sequencing-based contact maps provide global statistical organization, whereas imaging reveals single-cell structural heterogeneity.
8. Alternative cleavage and polyadenylation can place the 3' end of an mRNA at different positions. If a proximal poly(A) site is used upstream of sequences encoding

a transmembrane domain, the resulting mRNA may exclude that membrane-anchor coding region and produce a **secreted form**. If a distal poly(A) site is used, the transcript may retain the transmembrane segment and produce a **membrane-bound form**. Thus, cell-type-specific APA can alter both transcript structure and protein destination.